

Generative AI & LLM Systems Engineering

Duration: 3 Months (12 Weeks) | **Program Fee:** ₹2,999 INR | **Format:** 100% Practical Labs & Capstone Defense

Program Overview: Master enterprise-grade LLM architectures, production RAG pipelines, vector databases (Qdrant), PEFT/LoRA fine-tuning, and high-throughput vLLM serving.

3-MONTH STRUCTURED ROADMAP & MODULES

Module 01: LLM Foundations & Advanced Prompt Engineering

- Transformer decoder mechanics, self-attention & context windows
- Tokenization, temperature, top-p & sampling parameters
- Structured outputs, JSON mode & Chain-of-Thought (CoT) prompting

Lab Practical: *Build a production prompt-testing harness with schema validation in Pydantic.*

Module 02: Embeddings & Vector Database Pipelines

- Dense vs sparse embeddings (BGE, text-embedding-3, Cohere)
- Vector similarity metrics & indexing (HNSW, IVFFlat) in Qdrant
- Multi-source document chunking strategies & semantic boundaries

Architecture Pipeline: Document Parser → Semantic Chunker → Embedding Model → Qdrant Vector DB → Context Retriever

Module 03: Production RAG Architectures & Evaluation

- Hybrid search (BM25 + Dense vector) & cross-encoder rerankers
- Query transformation, hypothetical document embeddings (HyDE) & sub-queries
- Automated evaluation pipelines with RAGAS, TruLens & hallucination detection

Lab Practical: *Construct and benchmark an enterprise Q&A engine with automated faithfulness scoring.*

Module 04: Multimodal Vision & Audio Systems

- Vision-Language Models (VLM) with GPT-4o, Claude 3.5 & open-source Qwen2-VL
- Structured OCR document understanding & chart parsing
- Whisper audio transcription & real-time streaming speech pipelines

Module 05: LLM Fine-Tuning with PEFT & QLoRA

- Instruction tuning dataset preparation, formatting & synthetic data generation
- Parameter-Efficient Fine-Tuning (PEFT), LoRA & 4-bit QLoRA with Unsloth
- Training loss evaluation, over-fitting prevention & Hugging Face Hub deployment

Lab Practical: *Fine-tune an open-source LLM on custom technical documentation and publish adapters.*

Module 06: High-Throughput Model Serving & LLMOps

- Optimized inference with vLLM, PagedAttention & Ollama
- Streaming FastAPI endpoints & token throughput benchmarking
- Safety guardrails (NeMo Guardrails, Llama-Guard), latency monitoring & cost tracking

CAPSTONE PROJECT DEFENSE & DELIVERABLES

Title: Enterprise Multi-Tenant Knowledge RAG & Domain Fine-Tuned Copilot

Execution Pipeline: Document Ingestion → Hybrid Vector Index → Fine-Tuned Llama Adapter → vLLM Streaming API → Guardrails & Benchmarks

Deliverables: Production GitHub repo with modular clean code, Deployed high-throughput streaming FastAPI endpoint, Benchmark report comparing baseline vs domain-adapted RAG

Official curriculum issued by Unisole Skill AI Labs. Direct employer verification available via LinkedIn and student dashboard.